

backend/

```
|
|
|—— api-gateway/
| |—— src/
| | |—— config/
| | | |—— env.ts
| | |
| | |—— middleware/
| | | |—— auth.middleware.ts
| | | |—— rate-limit.middleware.ts
| | | |—— error.middleware.ts
| | | |—— request-logger.middleware.ts
| | |
| | |—— routes/
| | | |—— auth.routes.ts
| | | |—— user.routes.ts
| | | |—— product.routes.ts
| | | |—— inventory.routes.ts
| | | |—— order.routes.ts
| | | |—— payment.routes.ts
| | | |—— notification.routes.ts
| | |
| | |—— utils/
| | | |—— logger.ts
| | |
| | |—— app.ts
| | |—— server.ts
| |
| |—— tests/
| |—— Dockerfile
| |—— package.json
| |—— tsconfig.json
| |—— .env.example
|
|
|—— auth-service/
```

```
| |—— src/
| | |—— controllers/
| | |   └── auth.controller.ts
| | |
| | |—— services/
| | |   ├── auth.service.ts
| | |   ├── token.service.ts
| | |   ├── password.service.ts
| | |   └── verification.service.ts
| | |
| | |—— repositories/
| | |   └── auth.repository.ts
| | |
| | |—— routes/
| | |   └── auth.routes.ts
| | |
| | |—— middleware/
| | |   ├── authenticate.middleware.ts
| | |   └── authorize.middleware.ts
| | |
| | |—— validators/
| | |   └── auth.validator.ts
| | |
| | |—— events/
| | |   ├── publishers/
| | |   └── user-registered.publisher.ts
| | |
| | |—— config/
| | |   └── env.ts
| | |
| | |—— types/
| | |   └── auth.types.ts
| | |
| | |—— utils/
| | |   └── jwt.ts
| | |
| |—— app.ts
```

```
| | └── server.ts
| |
| |
| | └── openapi/
| |   └── auth.yaml
| |
| |
| | └── tests/
| | └── Dockerfile
| | └── package.json
| | └── tsconfig.json
| | └── .env.example
|
|
└── user-service/
    ├── src/
    |   ├── controllers/
    |   |   ├── user.controller.ts
    |   |   ├── address.controller.ts
    |   |   └── wishlist.controller.ts
    |   |
    |   ├── services/
    |   |   ├── user.service.ts
    |   |   ├── address.service.ts
    |   |   └── wishlist.service.ts
    |   |
    |   ├── repositories/
    |   |   ├── user.repository.ts
    |   |   ├── address.repository.ts
    |   |   └── wishlist.repository.ts
    |   |
    |   ├── routes/
    |   |   ├── user.routes.ts
    |   |   ├── address.routes.ts
    |   |   └── wishlist.routes.ts
    |   |
    |   ├── validators/
    |   ├── middleware/
    |   └── events/
```



```
| | | └── validators/
| | |   ├── product.validator.ts
| | |   └── category.validator.ts
| | |
| | | └── integrations/
| | |   └── clouinary.ts
| | |
| | | └── middleware/
| | | └── config/
| | | └── types/
| | | └── utils/
| | | └── app.ts
| | | └── server.ts
| | |
| | └── openapi/
| |   └── product.yaml
| |
| | └── tests/
| | └── Dockerfile
| | └── package.json
| | └── tsconfig.json
| | └── .env.example
| |
| |
| | └── inventory-service/
| |   ├── src/
| |   | ├── controllers/
| |   | | └── inventory.controller.ts
| |   |
| |   | └── services/
| |   |   └── inventory.service.ts
| |   |
| |   | └── repositories/
| |   |   └── inventory.repository.ts
| |   |
| |   | └── routes/
| |   |   └── inventory.routes.ts
```

```
| | |
| | |—— validators/
| | |  |—— inventory.validator.ts
| | |
| | |—— events/
| | |  |—— publishers/
| | |    |—— inventory-reserved.publisher.ts
| | |    |—— inventory-released.publisher.ts
| | |
| | |  |—— consumers/
| | |    |—— order-placed.consumer.ts
| | |    |—— order-cancelled.consumer.ts
| | |
| | |—— middleware/
| | |—— config/
| | |—— types/
| | |—— utils/
| | |—— app.ts
| | |—— server.ts
| |
| |—— openapi/
| |  |—— inventory.yaml
| |
| |—— tests/
| |—— Dockerfile
| |—— package.json
| |—— tsconfig.json
| |—— .env.example
|
|
|—— order-service/
|  |—— src/
|  |  |—— controllers/
|  |    |—— cart.controller.ts
|  |    |—— order.controller.ts
|  |
|  |
|  |—— services/
```

```
| | | |——— cart.service.ts
| | | |——— order.service.ts
| | |
| | | |——— repositories/
| | | |——— cart.repository.ts
| | | |——— order.repository.ts
| | |
| | | |——— routes/
| | | |——— cart.routes.ts
| | | |——— order.routes.ts
| | |
| | | |——— validators/
| | | |——— cart.validator.ts
| | | |——— order.validator.ts
| | |
| | | |——— events/
| | | |——— publishers/
| | | | |——— order-placed.publisher.ts
| | | | |——— order-cancelled.publisher.ts
| | | |
| | | |——— consumers/
| | | |——— inventory-reserved.consumer.ts
| | | |——— payment-confirmed.consumer.ts
| | | |——— payment-failed.consumer.ts
| | |
| | | |——— middleware/
| | | |——— config/
| | | |——— types/
| | | |——— utils/
| | | |——— app.ts
| | | |——— server.ts
| |
| |——— openapi/
| | |——— order.yaml
| |
|——— tests/
|——— Dockerfile
```

```
| |—— package.json
| |—— tsconfig.json
| |—— .env.example
|
|
|—— payment-service/
| |—— src/
| | |—— controllers/
| | | |—— payment.controller.ts
| | | |—— webhook.controller.ts
| | |
| | |—— services/
| | | |—— payment.service.ts
| | | |—— refund.service.ts
| | |
| | |—— repositories/
| | | |—— payment.repository.ts
| | |
| | |—— routes/
| | | |—— payment.routes.ts
| | | |—— webhook.routes.ts
| | |
| | |—— validators/
| | | |—— payment.validator.ts
| | |
| | |—— integrations/
| | | |—— stripe.ts
| | |
| | |—— events/
| | | |—— publishers/
| | | |—— payment-confirmed.publisher.ts
| | | |—— payment-failed.publisher.ts
| | | |—— refund-completed.publisher.ts
| | |
| | |—— middleware/
| | |—— config/
| | |—— types/
```



```
| | |—— utils/
| | |—— app.ts
| | |—— server.ts
| |
| |—— openapi/
| | |—— payment.yaml
| |
| |—— tests/
| |—— Dockerfile
| |—— package.json
| |—— tsconfig.json
| |—— .env.example
|
|
|—— notification-service/
| |—— src/
| | |—— controllers/
| | | |—— notification.controller.ts
| | |
| | |—— services/
| | | |—— notification.service.ts
| | | |—— email.service.ts
| | |
| | |—— repositories/
| | | |—— notification.repository.ts
| | |
| | |—— routes/
| | | |—— notification.routes.ts
| | |
| | |—— events/
| | | |—— consumers/
| | | |—— user-registered.consumer.ts
| | | |—— order-placed.consumer.ts
| | | |—— payment-confirmed.consumer.ts
| | | |—— payment-failed.consumer.ts
| | | |—— shipment-updated.consumer.ts
| | | |—— refund-completed.consumer.ts
```

```
| | |
| | |— templates/
| | | |— verification-email.ts
| | | |— password-reset.ts
| | | |— order-confirmation.ts
| | | |— payment-confirmation.ts
| | | |— payment-failed.ts
| | | |— shipment-update.ts
| | | |— refund-confirmation.ts
```

```
| | |
| | |— validators/
| | |— middleware/
| | |— config/
| | |— types/
| | |— utils/
| | |— app.ts
| | |— server.ts
```

```
| |
| |— openapi/
| | |— notification.yaml
```

```
| |
| |— tests/
| |— Dockerfile
| |— package.json
| |— tsconfig.json
| |— .env.example
```

```
|
|
|— shared/
| |— auth/
| | |— roles.ts
| | |— auth.types.ts
```

```
| |
| |— events/
| | |— event-types.ts
| | |— event-payloads.ts
```

```
| |
```

```
| |—— rabbitmq/
| | |—— connection.ts
| | |—— publisher.ts
| | |—— consumer.ts
| | |—— constants.ts
| |
| |—— errors/
| | |—— AppError.ts
| | |—— error-codes.ts
| |
| |—— types/
| |—— constants/
| |—— utils/
|
|—— package.json
|—— tsconfig.base.json
|—— .gitignore
|—— .env.example
|—— README.md
```

For our clothing e-commerce backend, we will build an API Gateway and seven core microservices: Auth, User, Product, Inventory, Order, Payment, and Notification.

Authentication will use JWT and refresh tokens with CUSTOMER and ADMIN roles. Product images will use Cloudinary, payments will use Stripe Test Mode, and RabbitMQ will handle required asynchronous events and notifications. Every service will expose health checks and documented REST APIs using OpenAPI/Swagger. The backend will provide all functionality required by the customer storefront and admin dashboard defined in the frontend implementation plan.